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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
11/917,269	02/29/2008	Shanker Kumar Mitra	066313 03/9-129	4561
909 7599 11/27/2013 Pillsbury Winthrop Shaw Pittman, LLP (NV) PO Box 10500 McLean, VA 22102				
			EXAMINER CLARK, AMY LYNN	
			ART UNIT 1658	PAPER NUMBER
			NOTIFICATION DATE 11/27/2013	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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heather.marthers@pillsburylaw.com

Office Action Summary	Application No. 11/917,269	Applicant(s) MITRA ET AL.	
	Examiner Amy L. Clark	Art Unit 1655	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.

- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
 - Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133).
- Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 05/05/2011.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election **was made** by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 1-4 and 6-29 is/are pending in the application.
5a) Of the above claim(s) 8-12, 14-18 and 29 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-4, 6, 7, 13 and 19-28 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some * c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☐ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date _____
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date: _____
- 4) ☐ Other: _____

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DETAILED ACTION

The present application is being examined under the pre-AIA first to invent provisions.

Acknowledgment is made of the receipt and entry of the amendment filed on 05/05/2011 with the cancellation of claim 5, and newly amended claims 1-4 and 13.

Election/Restrictions

The election/restriction requirement is maintained for the reasons of record.

Any rejection found in the previous Office Action and not repeated herein has been withdrawn based upon Applicant's amendments to the claims.

The text of those sections of Title 35, U.S. Code not included in this action can be found in a prior Office action.

Claims 1-4, 6, 7, 13 and 19-28 are currently under examination.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), first paragraph:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims 1-4, 6, 7, 13 and 19-28 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains



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subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for pre-AIA the inventor(s), at the time the application was filed, had possession of the claimed invention (newly applied as necessitated by amendment).

In the amended Claim 1, Applicant claims, "substantially solvent-free extract", thereby introducing a specific volume, which is considered to be new matter. Insertion of the above mentioned claim limitation has no support in the as-filed specification. The insertion of the limitation is a new concept because it neither has literal support in the as-filed specification by way of generic disclosure, nor are there specific examples of the newly limited genus which would show possession of the concept for a "substantially solvent-free extract". This is a matter of written description, not a question of what one of skill in the art would or would not have known.

The material within the four corners of the as-filed specification must lead to the generic concept. If it does not, the material is new matter. Declarations and new references cannot demonstrate the possession of a concept after the fact. Thus, the insertion of the above mentioned claim-limitation is considered to be the insertion of new matter for the above reasons.

As the above- mentioned claim limitation could not be found in the present specification, the recitation of the claim limitation is deemed new matter; and, therefore it must be omitted from the claim language, unless Applicant can particularly point to the specification for literal support.

Claims are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

The metes and bounds of claim 1 are rendered uncertain by the phrase "employing an organic solvent alone or mixture thereof and is soluble in" because it is unclear if Applicant is claiming that the extract is soluble in hexane, heptane, petroleum ether, etc or if Applicant is claiming that the organic solvent is soluble in these solvents. Also, what does Applicant mean by organic solvent or mixture thereof? Is Applicant claiming that the organic solvent can be mixed with other organic solvents or with

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water or something else? The lack of clarity renders the claims indefinite since the resulting claims do not clearly set forth the metes and bounds of the patent protection desired.

Claim Rejections - 35 USC § 102

Claims 1, 3, 4, 6, 7 and 13 remain rejected under 35 U.S.C. 102(b) as being anticipated by Singh et al. (Reference 1 on Applicant's IDS dated: 08/06/2010) (slightly modified based upon Applicant's amendments to the claims).

Singh teaches an extract of *Sphaeranthus indicus*, wherein the extract is obtained by extracting *Sphaeranthus indicus* flowers (which reads on aerial parts) with petroleum ether via Soxhalation and evaporated to dryness under reduced pressure (which reads on "substantially solvent-free", since removing solvent would make the extract free of solvent).

Although Singh does not teach that the extract of *Sphaeranthus indicus* is an anticancer agent, the claimed functional properties are inherent to the extract taught by Singh because the ingredients taught by Singh are one and the same as disclosed in the instantly claimed invention of Applicant. Thus, the extract of *Sphaeranthus indicus* taught by Singh is inherently an anticancer agent.

Therefore, the reference anticipates the claimed subject matter.

Claims 1, 3, 4, 6, 7, 13, 19, 20, 25 and 28 are rejected under 35 U.S.C. 102(b) as being anticipated by Bafna et al. (U).

Bafna teaches an extract of flower heads of *Sphaeranthus indicus* obtained by extraction with petroleum ether and combined with sodium carboxyl methyl cellulose (which reads on a pharmaceutically acceptable carrier) and injected into mice.

Although Bafna does not teach that the extract of *Sphaeranthus indicus* is an anticancer agent, the claimed functional properties are inherent to the extract taught by Bafna because the ingredients taught by Bafna are one and the same as disclosed in the instantly claimed invention of Applicant. Thus, the extract of *Sphaeranthus indicus* taught by Bafna is inherently an anticancer agent.

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Bafna does not expressly teach an extract of *Sphaeranthus indicus* obtained by percolation and/or soxhlation. However, it should be noted that Claim 13 constitutes a Product-by-Process type claim. In Product-by-Process type claims, the process of producing the product is given no patentable weight since it does not impart novelty to a product when the product is taught by the prior art. See *In re Thorpe*, 227 USPQ 964 (CAFC 1985); *In re Marosi*, 218 USPQ 289, 292-293 (CAFC 1983) and *In re Brown*, 173 USPQ 685 (CCPA 1972). Consequently, even if a particular process used to prepare a product is novel and unobvious over the prior art, the product *per se*, even when limited to the particular process, is unpatentable over the same product taught in by the prior art. See *In re King*, 107 F.2d 618, 620, 43 USPQ 400, 402 (CCPA 1939); *In re Merz*, 97 F.2d 599, 601, 38 USPQ 143-145 (CCPA 1938); *In re Bergy*, 563 F.2d 1031, 1035, 195 USPQ 344, 348 (CCPA 1977) *vacated* 438 US 902 (1978); and *United States v. Ciba-Geigy Corp.*, 508 F. Supp. 1157, 1171, 211 USPQ 529, 543 (DNJ 1979). Finally, since the Patent Office does not have the facilities for examining and comparing Applicant's composition with the compositions of the prior art reference, the burden is upon Applicant to show a distinction between the material, structural and functional characteristics of the claimed composition and the composition of the prior art. See *In re Best*, 562 F.2d 1252, 195 USPQ 430 (CCPA 1977).

Therefore, the reference anticipates the claimed subject matter.

Claims 1-4, 6, 7, 13, 19, 20 and 28 are rejected under 35 U.S.C. 102(b) as being anticipated by Kentebhrakam (V).

Kentebhrakam teaches a therapeutic formulation comprising the expressed juice of *Sphaeranthus indicus* bark, leaf, flower, fruit and root. Kentebhrakam further teaches that the formulation is an immunopotentiator. Kentebhrakam further teaches that the formulation can be combined with a medicinal liquid (which reads on a pharmaceutically acceptable carrier) and that the formulation can be orally administered.

Although Bafna does not teach that the extract of *Sphaeranthus indicus* is an anticancer agent, the claimed functional properties are inherent to the extract taught by Bafna because the ingredients

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taught by Bafna are one and the same as disclosed in the instantly claimed invention of Applicant. Thus, the extract of *Sphaeranthus indicus* taught by Bafna is inherently an anticancer agent.

Bafna does not expressly teach an extract of *Sphaeranthus indicus* obtained by an organic solvent or percolation and/or soxhlation. However, it should be noted that the claims constitute Product-by-Process type claims. In Product-by-Process type claims, the process of producing the product is given no patentable weight since it does not impart novelty to a product when the product is taught by the prior art. See *In re Thorpe*, 227 USPQ 964 (CAFC 1985); *In re Marosi*, 218 USPQ 289, 292-293 (CAFC 1983) and *In re Brown*, 173 USPQ 685 (CCPA 1972). Consequently, even if a particular process used to prepare a product is novel and unobvious over the prior art, the product *per se*, even when limited to the particular process, is unpatentable over the same product taught in by the prior art. See *In re King*, 107 F.2d 618, 620, 43 USPQ 400, 402 (CCPA 1939); *In re Merz*, 97 F.2d 599, 601, 38 USPQ 143-145 (CCPA 1938); *In re Bergy*, 563 F.2d 1031, 1035, 195 USPQ 344, 348 (CCPA 1977) *vacated* 438 US 902 (1978); and *United States v. Ciba-Geigy Corp.*, 508 F. Supp. 1157, 1171, 211 USPQ 529, 543 (DNJ 1979). Finally, since the Patent Office does not have the facilities for examining and comparing Applicant's composition with the compositions of the prior art reference, the burden is upon Applicant to show a distinction between the material, structural and functional characteristics of the claimed composition and the composition of the prior art. See *In re Best*, 562 F.2d 1252, 195 USPQ 430 (CCPA 1977).

Therefore, the reference anticipates the claimed subject matter.

Claim Rejections - 35 USC § 103

Claims 1, 3, 4, 13, 19-24 and 28 are rejected under 35 U.S.C. 103(a) as being unpatentable over Khanna (O, GB 2314269 A), in view of Kentebhrakam (V) and Batra et al. (P, WO 9961026 A1).

Khanna teaches a composition comprising an extract of all parts of *Sphaeranthus indicus* (which reads on all parts, flowers and aerial parts) in an amount of 1-5% (which reads on therapeutically effective amount) and lactose sugar, wherein the composition is formulated into a capsule (See pages 5-10). Khanna also teaches formulating an extract of *Sphaeranthus indicus* with sodium benzoate into a liquid (Which reads on an injectable form) and also formulating an extract of *Sphaeranthus indicus* with inert

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excipients into pellets, tablets or capsules for administration to patients. Khanna further teaches that the composition is useful for treating HIV/AIDS.

The teachings of Kentebhrakam are set forth above and applied as before.

Batra teaches a capsule and tablet formulation for treating HIV infections and AIDS comprising microcrystalline cellulose (which is synonymous with MCCP IP), magnesium stearate, pregelatinized starch, colloidal silicon dioxide, croscarmellose sodium, crospovidone, and water.

It would have been obvious to modify the composition taught by Khanna by using expressed juice of *Sphaeranthus indicus* as the extract as taught by Kentebhrakam with microcrystalline cellulose (which is synonymous with MCCP IP), magnesium stearate, pregelatinized starch, colloidal silicon dioxide, croscarmellose sodium, crospovidone, and water, as taught by Batra because at the time the invention was made, it was known that an expressed juice extract of *Sphaeranthus indicus*, microcrystalline cellulose (which is synonymous with MCCP IP), magnesium stearate, pregelatinized starch, colloidal silicon dioxide, croscarmellose sodium, crospovidone, and water were all useful ingredients that could be administered orally for treating HIV/AIDS in the form of a tablet or capsule, as clearly taught by the above references.

Thus, an artisan of ordinary skill would reasonably expect that substituting expressed juice of *Sphaeranthus indicus* taught by Kentebhrakam for the extract taught by Khanna and combining an extract of *Sphaeranthus indicus*, microcrystalline cellulose, magnesium stearate, pregelatinized starch, colloidal silicon dioxide, croscarmellose sodium, crospovidone, and water would provide an even more effective medicine for treating HIV/AIDS and would provide a more effective delivery system for an extract of *Sphaeranthus indicus*. This reasonable expectation of success would motivate the artisan to use expressed juice of *Sphaeranthus indicus*, microcrystalline cellulose, magnesium stearate, pregelatinized starch, colloidal silicon dioxide, croscarmellose sodium, crospovidone, and water in a formulation for treating HIV/AIDS based upon the teachings of the above references.

The result-effective adjustment of particular conventional working conditions is deemed merely a matter of judicious selection and routine optimization which is well within the purview of the skilled artisan.

Based upon the beneficial teachings of the cited references, the skill of one of ordinary skill in the

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art, and absent evidence to the contrary, there would have been a reasonable expectation of success to result in the claimed invention.

Accordingly, the claimed invention was prima facie obvious to one of ordinary skill in the art at the time the invention was made, especially in the absence of evidence to the contrary.

Claims 1, 3-7, 13, 19-21, 23 and 25-28 are rejected under 35 U.S.C. 103(a) as being unpatentable over Khanna (O, GB 2314269 A), in view of Kentebhrakam (V) and Kalden et al. (A, 5,334,612).

The teachings of Khanna and Kentebhraka are set forth above and applied as before.

Kalden teaches a composition for treating retrovirus diseases, wherein the composition can be injected and wherein the composition can comprise castor oil, ethanol or water and benzyl alcohol (as a solvent but which also known to be an effective preservative).

It would have been obvious to modify the composition taught by Khanna by substituting the extract of *Sphaeranthus indicus* with pressed juice of *Sphaeranthus indicus* and combining an extract of *Sphaeranthus indicus* with castor oil, ethanol or water and benzyl alcohol, as taught by Kalden. At the time the invention was made, it was known that an extract of *Sphaeranthus indicus*, castor oil, ethanol or water and benzyl alcohol were all useful ingredients that could be administered in the form of a liquid and could be injected (please note that liquids that are safe for ingestion can be injected), as clearly taught by the above references.

Thus, an artisan of ordinary skill would reasonably expect that combining an extract of *Sphaeranthus indicus*, castor oil, ethanol or water and benzyl alcohol would provide an even more effective delivery system for an extract of *Sphaeranthus indicus*. This reasonable expectation of success would motivate the artisan to use an extract of *Sphaeranthus indicus*, castor oil, ethanol or water and benzyl alcohol in a formulation for treating HIV/AIDS based upon the teachings of the above references.

The result-effective adjustment of particular conventional working conditions (e.g., determining the purity of water and the purity of alcohol, as well as a suitable form of castor oil) is deemed merely a matter of judicious selection and routine optimization which is well within the purview of the skilled artisan.

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Based upon the beneficial teachings of the cited references, the skill of one of ordinary skill in the art, and absent evidence to the contrary, there would have been a reasonable expectation of success to result in the claimed invention.

Accordingly, the claimed invention was prima facie obvious to one of ordinary skill in the art at the time the invention was made, especially in the absence of evidence to the contrary.

Response to Arguments

Applicant's arguments filed 05/05/2011 have been fully considered but they are not persuasive. Applicant argues that the product taught by Singh is eluted by hexane, petroleum ether and benzene successively to remove fatty material, carotenoids and phytosterols from the gel and that the remaining material on the silica gel is eluted by ethyl acetate and that this is Singh's product. However, this is not found persuasive because Singh clearly teaches extracting *Sphaeranthus indicus* with the instantly claimed solvents and that the products recovered from the column prior to eluting with ethyl alcohol are those claimed by Applicant (since there would be residual solvent left on the column, and elution with the solvents taught by Singh that are identical to those claimed by Applicant would provide the instantly claimed product prior to elution with ethyl acetate). Whether Singh intended this elution to be the final product or not, the solvent extract obtained by elution as taught by Singh is identical to that claimed by Applicant. Finally, Singh teaches a primary extraction with an instantly claimed solvent, so even prior to purification, the instantly claimed extract is taught by Singh. Therefore the rejection is maintained for the reasons of record and the reasons set forth herein.

Applicant's arguments with respect to claims 1-4, 6, 7, 13 and 19-28 have been considered but are moot because the arguments do not apply to any of the references being used in the current rejection.

Conclusion



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No claims are allowed.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Amy L. Clark whose telephone number is (571)272-1310. The examiner can normally be reached on Monday to Friday between 7 am - 3:30 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Terry McKelvey can be reached on (571) 272-0775. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.



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/Amy L Clark/

Primary Examiner, Art Unit 1655

Notice of References Cited	Application/Control No. 11/917,269	Applicant(s)/Patent Under Reexamination MITRA ET AL.	
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U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number Kind Code	Date MM-YYYY	Name	Classification
	A	US-			
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
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	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	Bafna et al. Ars Pharmaceutica, 45:3 (2004) 281-291.
	V	Kentebhrakam (02) From "TKDL" AB/971. Known for 500 years. Prior art citation at bottom of page: August 1999. Retrieved on: 2013-11-22 From TKDL website.
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.02(a)).
Dates in MM/YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 11917269	Applicant(s)/Patent Under Reexamination MITRA ET AL.
	Examiner AMY L CLARK	Art Unit 1655

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
NONE	NONE		

SEARCH NOTES		
Search Notes	Date	Examiner
EAST- See Search Notes	11/22/2013	ALC
Inventors names searched in PALM	11/22/2013	ALC
Google/TKDL search for terms including sphaeranthus indicus and synonyms	11/22/2013	ALC

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
NONE	NONE		

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